Work Order NO :- OPRL/ATE/11094106758
February 14, 2025
Revision Number :- 0

MARCO ENTERPRISES (PWM875)
PNB BANK, KHASRA NO-237, MAIN ROAD HAZIPUR, Sector 104, Noida, Gautam Buddha Nagar, UP

GST No.	**09BLDPK5228H1Z2**	**EPF No.**
PAN No.	**BLDPK5228H**	**ESI No.**
		Labour License No.

Project :-	**Omaxe Ananda Tower-E at Prayagraj, U.P. (127018)**
Subject :-	**Work Order for Civil Structure Work (Location: Tower-E Stilt + 14 Floors)**
Validity of Contract:-	**Upto 31-08-2026 (18 Months)**
Kind Attn:-	**Mr. Mohit Rana (9910041120) (Email id:- manish.sharma@marcogroup.in)**

Dear Sir,

This has reference to the discussions and negotiations we had with you, we are pleased to place the work order for the above said work amounting to **Rs.16,20,74,375/- (Rupees Sixteen Crore Twenty Lakhs Seventy Four Thousand Three Hundred Seventy Five Only excluding GST)** as per following rates, terms & conditions and specification.

S. No.	Description	Unit	Qty	Rate	Amount
1.	**SAC Code:** **General Notes -** The following notes shall apply to all Sections and Bills contained in this Bill of Quantities: 1. The quantities, classes and kinds of work set out in this Bill of Quantities are provisional. They do not purport to or represent the final quantities, classes or kinds of work eventually required to be done. 2. This Bill of Quantities is not intended for and shall not be used for ordering purposes. 3. This Bill of Quantities is to be read in conjunction with the Drawings, Specifications & Preambles to Work Sections which shall be construed as being complementary to one another. Any discrepancy shall be dealt in accordance with the Conditions of Contract. 4. Works shall be executed at all heights / depths, levels, floors and locations all complete as per Drawings, Specifications and to the complete satisfaction of the Engineer-In-Charge. 5. Unless otherwise specified, mode and method of measurement shall be in accordance with				

	IS:1200. 6. Contractor should strictly follow the Site Safety Manual and provide & install all the safety measures required as per Manual. 7. The description of an item shall be deemed to include all preambles, headings, sub-headings and references to other documents. The detailed description of work and materials, given in the Specification are not necessarily repeated in the Bill of Quantities. 8. The Preamble to the Bills of Quantities must be read with the Drawings, Conditions of Contract and the Specification and the tenderer shall be deemed to have examined the Drawings, Specifications, Conditions of Contract and to have acquainted himself with the Works to be done and the way in which they are to be carried out. This preamble to the Bills of Quantities shall form part of the Contract. 9. Providing and maintaining Field Laboratory in a satisfactory manner till the completion of work to ensure site control on quality of materials and the works and shall strictly comply with provision of Indian/ International Standards at no extra cost. All other regular tests including requisite materials shall be carried out as directed either at approved laboratory or at site at no extra cost. 10. The Tenderers are requested to inspect the site and obtain for themselves at their own expense all necessary information and particulars to enable them to submit the tender. 11. The dimensions shown in the drawings shall have to be verified by the contractor at site before execution of the items and necessary allowance's have to be made as per the actual dimensions available at site. 12. For the items of works having Basic rate of material, price adjustment will be made only on the actual quantity measured. Wastage shall be to Contractors account. Basic rate indicates landed cost at site including transportation charges, loading-unloading as applicable. 13. The contractor should refer to the detailed drawings for details. All the rates so quoted should be inclusive of all hardware. In case of discrepancy between drawing and Bill of Quantity the drawing supersedes, unless specifically mentioned in the BOQ. 14. The work shall be carried out as per				

	drawings, specifications, the description of the items in the schedule and / or engineer's instructions. 15. All pre-fabrication works wherever possible shall be carried outside the client's premises, only assembling and finishing work will be allowed within the Site. 16. Items of work provided in this schedule but not covered in the specifications shall be executed strictly as per instructions of the engineer. 17. Unless otherwise mentioned in the contract, the tenderer shall quote for the finished items and shall provide for the complete cost towards labour, materials, plant and machinery, operational costs, levies, taxes, transport, repairs, rectification, maintenance till handing over, revenue expenses, contingencies, overheads, profits and all incidental items not specifically mentioned but reasonably implied and necessary to complete the works according to the contract. 18. The quantities of the various items mentioned in the schedule of items are approximate and may vary up to any extent or be deleted altogether. The contractor, in his own interest, should get an indication of the probable extent of the work to be executed under any particular item in this schedule, before undertaking any preliminary and enabling work or purchasing bought out components related to the work. 19. The Engineer-in-charge's decision shall be final and binding on the contractor regarding clarifications of items in this schedule with respect to the other sections of the Contract. 20. It may be noted that the total work or any part / parts of the same may be awarded as mentioned in this schedule of items. 21. The rates quoted shall be inclusive of all mock-ups, associated samples prior to any mass production or procurement as per complete satisfaction of the Client and Architect. Inclusive of dismantling, demolishing and carting the material and making good of the surface. 22. Any debris generated out of the said work may please be cleared on priority basis from the Site. The contractor to allow for work to be executed on the upper Floors, temporary storage of material on the Ground Floor, for hoisting material to the upper floors, for bringing debris and waste down, for co-ordinating and				

	scheduling movement of material with facilities available on site. The disposal of debris will be the sole responsibility of the Contractor. 23. Any fault that occurs during the time of production, shifting or installation of finished materials should be replaced with good one as per the instruction from Engineer-in-charge. 24. Maintaining the barricade and repair if required by the contractor. 25. Making lay out with help of professional/In house survey agency, establishing bench marks, permanent grid lines on permanent objects and checking verticality of entire structure floor wise at regular intervals, and producing drawings, and all survey works related to project, using Total Station, laser beam device. Laser beam range of minimum 200 metres to be used for checking of verticality of entire structure. 26. Protection- Cover up and protect from damage, including inclement weather including all works and materials but not limited to , all finished works, unfixed materials, to the satisfaction of the Project Manager. 27. Keeping the site clear, clean, and meeting the hygiene standards- Cost of providing and maintaining the workers toilets shall be covered in the scope of contractor. As per safety norms mentioned elsewhere.				
1.1.	**SAC Code:** **EARTH WORK:**				
1.1.1.	**SAC Code:** Earth work in excavation in foundation trenches, column pits, plinth beam, under grade/ raft slab etc. by mechanical and/or manual means up to a depth of 1.5 mtr in soil from existing excavated soil level, stacking/spreading the excavated material within site premises (within a stretch of 1500 M) as per direction of Project Manager including dressing of bottom, sides strutting, timbering, shoring, trimming of sides, all leads and lifts, bailing or pumping out the water from the excavated areas collected from any source including subsoil water and keeping the excavated surface dry for subsequent works incidentals, etc. complete with ramming of bottom, surface preparation as per specifications to receive the PCC.	CUM	1000	120.00	120,000.00

	Excavation shall be payable only for vertical cut up to the outer edge of foundation lean concrete as indicated in drawings for all depths. Cost of extra excavation required for working space, supporting side shuttering of foundations and retaining walls, planking, dewatering, strutting etc. and that required for making slopes for stability of excavation shall be deemed to be included in the rate quoted. **This item is not for Mass Excavation.**				
1.1.2.	**SAC Code:** Prepare the surface for PCC by manual excavation upto 30 cm depth including dressing the surface, ramming the bottom, disposal and spreading of the surplus earth at the specified location within premises including proper levelling, watering and consolidation complete as directed by E I C. (The item shall be measured as per plan area of PCC, Working space shall not be taken into measurement, deemed to be included into rates) **(This item is only applicable if excavation item is not being executed by contractor and excavation already done by client for the location)**	SQM	1500	65.00	97,500.00
1.1.3.	**SAC Code:** Filling available excavated earth in trenches, plinth, sides of foundations etc. in layers not exceeding 20cm in depth: consolidating each deposited layer by ramming and watering, lead within the premises and upto any depth. **(Compacted Volume shall be payable as per drawing) - Actual volume shall be paid incase excavation is done by Client.**	CUM	2400	100.00	240,000.00
2.	**SAC Code:** **CONCRETE WORK:** Rates for all cement concrete items plain and reinforced described below shall include work from Basement Level to top of last roof including roof of machine room, mumty, over head water tanks and parapets above at all floors and for all heights. The contractor should provide full time sufficient pipes with accessories and pumps based on the site requirement for pumping and laying of concrete as directed by the EIC. The rate inclusive of standard rigid pipe line and its supports, lifting, laying charges, etc., at all				

	heights and levels. No additional payment shall be payable to the contractor on account of using or change in any size of coarse aggregate in PCC/ RCC work at any stage during currency of works. **(This section covers cast in situ structural and architectural as well.)**				
2.1.	**SAC Code:** **Plain Cement Concrete:** Providing and laying plain cement concrete of specified grade as bed concrete under foundation, bases for rafts, underground water tank, building machine equipments, column footing, under floor landings, below road and pathways, cill, coping, screeding at roof to required slope and other locations as called for laid, consolidated and cured etc. complete including finishing the top surface to receive the next course / layer as per specification and drawing. **(Shuttering shall be paid in Item No.3.1.1)** **M10 Grade (Basic Rate of PCC - Rs. 4200/cum without pump, without GST F.O.R site)**	CUM	100	5,550.00	555,000.00
2.2.	**SAC Code:** Providing and laying cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20mm nominal size) for ledge wall below balcony railings, band, mullion, lintel etc. **(Shuttering & reinforcement shall be paid separately in respective item)** as per drawing and as directed by the Project Manager.	CUM	50	5,750.00	287,500.00
2.3.	**SAC Code:** Providing and laying damp proof course 40mm thick including shuttering with cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20mm nominal size) including mixing of water proofing compound @ 2% of cement content and applying two layers of bitumen @ 1.70 Kg/Sqm of grade 85/25 and as directed by the Project Manager.	SQM	200	480.00	96,000.00
2.4.	**SAC Code:** **RCC WORK:** Providing & Laying RMC M-25 Grade as per approved design mix concrete (strictly controlled conditions) with flyash (as per approved design mix) with automatic batching plant machine mixed and machine vibrated RCC with melamine based plasticisers etc., upto any level, including finishing the exposed surface,				

	fixing/ removal of pockets, fixing of sleeves and inserts (sleeves provided by client), curing etc complete, as per drawing (including all size and shape like circular, semi-circular, oval etc.) and as directed for all lead and lift, excluding cost of centering, shuttering and steel reinforcement but including cost of making shear keys and provision of construction / contraction / control joints in various locations. Checking from Client should not be considered as final and any defects arising during or after casting should be borne & bear by contractor at his own cost. This is inclusive of leaving shuttering & centring for pour strip, slab in slope, RCC members in curved, etc., nothing is payable extra on this account. **RMC (M-25) Basic Rate - Rs. 5200/cum without pump excluding GST**				
2.4.1.	**SAC Code:** Foundation, column footings, raft slab, raft beams, plinth beam, lift pits, sump pits, pedestals / bases of column, column upto plinth, swimming pool, Retaining wall foundations, grade slab etc.	CUM	2080	6,600.00	13,728,000.00
2.4.2.	**SAC Code:** Suspended floors, roofs, balconies, shelves/ counters, stair waist slabs, folded plate stairs and landings, swimming pool slab, lintel and floor beams, landing beams, girders (deep beams), RCC bands, swimming pool beam, cantilever projections, coping, cills, slab of tank etc at all levels.				
2.4.2.1.	**SAC Code:** Stilt Floor level to 7th Floor level	CUM	2200	6,550.00	14,410,000.00
2.4.2.2.	**SAC Code:** 7th Floor level to to Terrace Including OHT, Mumty, Machine Room, Parapet	CUM	2440	6,750.00	16,470,000.00
2.4.2.3.	**SAC Code:** All type of Moulding, Bands, Elevation/Architectural Features as per drawing and as directed by EIC. at all level	CUM	70	7,300.00	511,000.00
2.5.	**SAC Code:** Columns, Wall, pillars, piers, abutments, posts and struts, offset and projection of columns,				

	water tank walls, lift walls, Retaining Wall, swimming pool, Parapets, planters, vertical projection of pergolas, wall and ramp walls etc. at all levels.				
2.5.1.	**SAC Code:** Stilt Floor level to 7th Floor level	CUM	860	6,850.00	5,891,000.00
2.5.2.	**SAC Code:** 7th Floor level to to Terrace Including OHT, Mumty, Machine Room, Parapet	CUM	950	7,050.00	6,697,500.00
2.5.3.	**SAC Code:** All type of Moulding, Bands, Elevation/Architectural Features as per drawing and as directed by EIC. at all level	CUM	30	7,650.00	229,500.00
2.54.	**SAC Code:** Finishing vaccum dewatering continuous concrete flooring 100-150 mm thick with specified grade concrete including necessary formwork in required level, vibrating concrete both by poker vibrators / leveling vaccum processing of the concrete and planning the surface by skin powered floater trowels with broom/plain finish including groove cutting & filling etc. complete. **(Concrete shall be paid in the respective items above).**	SQM	1625	100.00	162,500.00
3.	**SAC Code:** **CENTRING AND SHUTTERING:**				
3.1.	**SAC Code:** Providing and fixing plywood or steel plate shuttering and centring for structural concrete work in locations called for including strutting, propping, bracing, bolting, wedging, casing, striking, removal etc. complete for any size, section, thickness. Props shall consist of well designed steel pipes adequately braced (wooden ballies as props shall not be permitted). The item to include centering and shuttering at all heights and levels. Allow for forming grooves, drip course, chamfers, cut-outs, openings etc. where called for and for dressing with approved shuttering oil to prevent adhesion. Unexposed concrete surfaces subsequently left untreated (in the condition obtained on removal of form work) shall not constitute exposed concrete work and shall be measured under this item only Irregular				

	shapes with sides (like circular, semi-circular, oval etc.) which may not be perpendicular to each other for shapes not conforming to rectangle or square, and circular shuttering shall not be paid separately. Joints of soffits of the slab shall be made clean after concreting and de-shuttering. **(quoted rate to include all staging upto 3.50 mtr height).** **Note:- Contractor shall take necessary precautions such as watertight shuttering. This is inclusive of RCC members, leaving shuttering & centring for pour strip, slab in slope, RCC members in curved (in plan), etc., nothing is payable extra on this account.**				
3.1.1.	**SAC Code:** Foundation, column footings, raft slab, raft beams, plinth beam, lift pits, sump pits, pedestals / bases of column, column up to plinth, swimming pool, Retaining wall foundations, grade slab etc.	SQM	2150	360.00	774,000.00
3.1.2.	**SAC Code:** Suspended floors, roofs, balconies, shelves/ counters, stair waist slabs, folded plate stairs and landings, swimming pool slab, lintel and floor beams, landing beams, girders (deep beams), RCC bands, swimming pool beam, cantilever projections, coping, cills, slab of tank etc at all levels.				
3.1.2.1.	**SAC Code:** Stilt Floor level to 7th Floor level	SQM	17600	465.00	8,184,000.00
3.1.2.2.	**SAC Code:** 7th Floor level to to Terrace Including OHT, Mumty, Machine Room, Parapet	SQM	18800	485.00	9,118,000.00
3.1.2.3.	**SAC Code:** All type of Moulding, Bands, Elevation/Architectural Features as per drawing and as directed by EIC. at all level	SQM	400	550.00	220,000.00
3.2.	**SAC Code:** Columns, Wall, pillars, piers, abutments, posts and struts, offset and projection of columns, water tank walls, lift walls, Retaining Wall, swimming pool, Parapets, planters, vertical projection of pergolas, wall and ramp walls etc. at all levels.				

3.2.1.	**SAC Code:** Stilt Floor level to 7th Floor level	SQM	7825	465.00	3,638,625.00
3.2.2.	**SAC Code:** 7th Floor level to to Terrace Including OHT, Mumty, Machine Room, Parapet	SQM	8200	485.00	3,977,000.00
3.2.3.	**SAC Code:** All type of Moulding, Bands, Elevation/Architectural Features as per drawing and as directed by EIC. at all level	SQM	150	550.00	82,500.00
3.3.	**SAC Code:** Extra for additional height in centring, shuttering where ever required with adequate bracing, propping etc. including cost of de-shuttering and decentring at all levels, over and above height of 3.50 mtr **(Plan area to be measured and shall be paid proportionately on actual height) - for every 1.0 mtr. height**	SQM	1000	115.00	115,000.00
4.	**SAC Code:** **REINFORCEMENT:**				
4.1.	**SAC Code:** Receiving, laying and fixing in position steel reinforcement in all reinforced concrete work, including straightening, cutting, removal of loose rust by wire brush and coating with cement slurry, bending, hoisting, laying in position to the shape and profile required at all levels and heights as per drawing and design and/ or as directed, binding with double folded steel wire as approved by EIC etc. complete. (Contractor shall prepare BBS and submit to EIC for approval prior to cutting, bending & laying of steel bars and payments shall be released as per approved BBS complete in all respect as per direction and approval of EIC Rate shall include cost of 18 gauge GI binding wire, cover blocks as per Indian Standards, unloading the steel & moving to the steel yard, moving site to site within project etc) (Only Steel reinforcement shall be provided by the owner free of cost at one point) Thermomechanically Treated Reinforcement (TMT) or High yield strength Deformed (Tor				

	steel) bars of all size & grades.				
4.1.1.	**SAC Code:** Up to Stilt Floor Level	MT	150	8,500.00	1,275,000.00
4.1.2.	**SAC Code:** Stilt Floor level to to Terrace Including OHT, Mumty, Machine Room, Parapet)	MT	805	9,400.00	7,567,000.00
4.1.3.	**SAC Code:** All type of Moulding, Bands, Elevation/Architectural Features as per drawing and as directed by EIC. at all level	MT	10	11,000.00	110,000.00
5.	**SAC Code:** **MASONARY WORK:** (Quoted Rates are for all heights, all depths, all levels, including basement to roof (mumty) & all leads and lifts, including curved or circular including elevation masonary work) **(Basic Rate of Brick @ Rs. 7/No. FOR at site without GST)**				
5.1.	**SAC Code:** **Brick work in Sub Structure (Up to Stilt Floor level):** Providing and laying 230mm thick brick masonry using 1st quality burnt clay FPS bricks in Sub-Structure laid in cement mortar 1:6 (1 cement : 6 coarse sand) mix, joints finished, flush/ raked to 6mm depth including scaffolding, curing complete as per specification and drawing or as directed by Project Manager.	CUM	100	6,000.00	600,000.00
5.2.	**SAC Code:** **Brick work in Sub Structure (Up to Stilt Floor level):** Providing and laying of 115 mm thick brick work with 1st quality burnt clay FPS bricks in sub-superstructure laid in cement mortar 1:4 (1 cement : 4 coarse sand) mix, joints finished, including placing hoop iron as per specification at every third course, scaffolding, curing complete in all respect as per direction of EIC. **(Hoop iron shall be provided by client free of cost) (Items of RCC bands like shuttering, concrete and reinforcement to be paid in respective items).**	SQM	1000	800.00	800,000.00
5.3.	**SAC Code:** **Brick work in Super Structure (Above Stilt Floor level):** Providing and laying 230mm thick brick masonry using 1st quality burnt clay FPS				

	bricks in Super-Structure laid in cement mortar 1:6 (1 cement : 6 coarse sand) mix, joints finished, flush/ raked to 6mm depth including scaffolding, curing complete as per specification and drawing or as directed by Project Manager.				
5.3.1.	**SAC Code:** Stilt Floor level to 7th Floor level	CUM	1185	6,300.00	7,465,500.00
5.3.2.	**SAC Code:** 7th Floor level to to Terrace Including OHT, Mumty, Machine Room, Parapet	CUM	1355	6,500.00	8,807,500.00
5.4.	**SAC Code:** **Brick work in Super Structure (Above Stilt Floor level):** Providing and laying of 115 mm thick brick work with 1st quality burnt clay FPS bricks in Super-Superstructure laid in cement mortar 1:4 (1 cement : 4 coarse sand) mix, joints finished, including placing hoop iron as per specification at every third course, scaffolding, curing complete in all respect as per direction of EIC. **(Hoop iron shall be provided by client free of cost) (Items of RCC bands like shuttering, concrete and reinforcement to be paid in respective items).**				
5.4.1.	**SAC Code:** Stilt Floor level to 7th Floor level	SQM	7100	875.00	6,212,500.00
5.4.2.	**SAC Code:** 7th Floor level to to Terrace Including OHT, Mumty, Machine Room, Parapet	SQM	8155	915.00	7,461,825.00
5.5.	**SAC Code:** **Brick work in staircase steps:** Providing and laying and constructing brickwork in staircase steps using 1st quality burnt clay FPS bricks in cement mortar 1:3 (1 cement : 3 coarse sand) at all heights and levels including cutting, chamfering scaffolding, curing, complete as per specification.	CUM	40	7,000.00	280,000.00
6.	**SAC Code:** **FINISHING WORKS (PLASTER):** (Quoted Rate shall be for all heights, depths, levels, leads and lifts and to include providing and fixing chicken wire mesh in junction RCC and brick / AAC Block walls and at all chased for electrical conduits & plumbing pipes etc. for				

	internal and external plaster works including minor chipping of RCC, brickwork surface).				
6.1.	**SAC Code:** **Internal Plaster Super-Structure:** **Wall Plaster:** Providing and applying 12 to 15mm thick plaster to masonry and RCC walls, columns etc. in cement mortar 1:5 (1 cement : 5 screened coarse sand) finished smooth including hacking the surfaces, steel scaffolding, curing, making grooves at desired location etc. complete for all levels as per approved drawing.				
6.1.1.	**SAC Code:** Stilt Floor level to 7th Floor level	SQM	19970	265.00	5,292,050.00
6.1.2.	**SAC Code:** 7th Floor level to to Terrace Including OHT, Mumty, Machine Room, Parapet	SQM	22825	280.00	6,391,000.00
6.2.	**SAC Code:** **Internal Plaster Super-Structure:** **Ceiling Plaster:** Providing and applying 6-10mm thick plaster to RCC slab, beam in cement mortar 1:4 (1 cement : 4 screened coarse sand) finished smooth including hacking the surfaces, steel scaffolding, curing, making grooves at desired location etc. complete for all levels as per approved drawing.				
6.2.1.	**SAC Code:** Stilt Floor level to 7th Floor level	SQM	9135	255.00	2,329,425.00
6.2.2.	**SAC Code:** 7th Floor level to to Terrace Including OHT, Mumty, Machine Room, Parapet	SQM	10440	270.00	2,818,800.00
6.3.	**SAC Code:** **Double Layer External Plaster:** Providing and applying 20 mm thick external cement plaster in two coats under layer 12 mm thick cement plaster 1:5 (1 cement : 5 screened coarse sand) finished with a top layer 8mm thick cement plaster 1:4 (1 cement : 4 screened coarse sand) including making of grooves, bands, drip coarse, design etc. as shown in drawing, steel scaffolding, curing, providing & mixing water proofing compound (CICO) in both layers as per manufacturer specification etc. complete as per approved drawing.	SQM	19200	480.00	9,216,000.00

6.4.	**SAC Code:** **Single Layer (Shaft Plaster):** Providing and applying 12-15 mm thick external cement plaster in single coat finished with cement plaster 1:4 (1 cement : 4 screened coarse sand) including making of grooves, bands, drip coarse as shown in drawing, steel scaffolding, curing, providing & mixing water proofing compound (CICO) as per manufacturer specification etc. complete as per approved drawing.	SQM	1000	380.00	380,000.00
7.	**SAC Code:** **STRUCTURE STEEL:** NOTES: The rates quoted by contractor for items under this head shall be inclusive of the following:- (i) Fabrication, cutting, bending to required size and shape, drilling, punching, counter-sinking, welding, grinding, filling etc. finishing, erecting and fixing in position at all heights and depths with all leads and lift. (ii) M.S. ladders and MS Railings in staircases, balconies and at any other place etc shall be fixed by welding the railing to steel reinforcement of slabs/beams/columns/staircase etc to make it absolutely rigid. The rate includes cutting of RCC required if any and making good the same after welding of the railing. Subsequently, the same shall be embedded in cement concrete of specified grade. (iii) M.S. grills/weld mesh etc. shall be fixed to masonry/concrete jambs of floor by means of required size and number of M.S. lugs grouted/ embeded in cement concrete block 1:2:4 or fixed by means of suitable counter sunk screw with rawal plugs or fixed by dash fastener of required dia and depth. (iv) The accessories such as M.S. hold-fasts / dash fasteners, lugs screws, rawal plug, G.I. pipe pieces, grouting in C.C. block 1:2:4 etc. shall not be paid separately. This shall be included in rate of items. (v) All steel work shall be painted with one coat of red oxide primer of approved make. (vi) No extra shall be paid for transport of fabricated items from one place to another place. (vii) Cost of all erection fasteners shall be borne by the contractor.				

	(viii) All Welding work should be done as per IS:814. (ix) Domestic LPG cylinder shall not be used for any cutting of steel members. (x) All holes shall be drilled and no gas cutting shall be Allowed. (xi) Localized 'earthing' to be provided while welding to avoid any damage to electronic equipments in the running plant.				
7.1.	**SAC Code:** Structural steel work with solid/hollow sections: Providing and fixing M.S. work in built-up section such as staircase railing, balcony railing, grating, MS drain cover, MS mumty door including hold fast/fastener etc. with square/round MS Pipes, MS flats, MS Hollow section etc., as per design electrically welded and fixing in RCC/brick masonry or/and treads of steps with cement concrete 1:2:4 (1 cement : 2 corse sand : 4 grades stone aggregate 20 mm nominal sie), complete in all respect as per drawing & direction of EIC, including a coat of red oxide steel primer. **(Basic Rate of Str. Steel - Rs.60/- Per Kg. F.O.R excluding GST)**	KG	39665	110.00	4,363,150.00
8.	**SAC Code:** **WOOD WORK:** NOTES: The rates quoted by contractor for items under this head shall be inclusive of the following:- i) All labour, materials, use of tools and equipment to complete the Wood works with all leads and lift. ii) The rate shall be inclusive for applying water proof wood primer and two coats of termite proof paint on concelled surfaces on the wooden frame. iii) Nails, Glue, Screws, Dash Fastener, lipping, Door hardware i.e. Mortice Lock, Handle, Tower bolt, Door Stopper, Door Buffer, Magic eye etc. are included in the rate. iv) All joints shall be approved by the Client and according to the best practice. All joinery work shall be glued with best quality synthetic waterproof adhesive Fevicol or equivalent make as approved. v) All screws shall be driven with screwdriver and shall not be hammered. vi) Quoted rates are applicable for all depths,				

	heights, lead & lift unless specified explicitly.				
8.1.	**SAC Code:** Providing and Fixing seasoned Red Meranti door frame including painting the portion in contact with masonary / R.C.C. with 2 or more coats of Antitermite treatment paint and fixng the frames with suitable 3 Nos. M.S. holdfasts each side made out of 40 x 5 mm flat iron 230 mm long with bolts and nuts/ dash fasteners for RCC surfaces, duly embedded in C.C. Blocks of required size in (1:2:4) 1 cement : 2 coarse sand : 4 stone aggregate 20 mm nominal size) including tarring the hold fast etc.all complete as per direction of EIC. **(Finished size to be confirmed by client before procurement) (Sample of wood shall be get approved by client before procurement)**	CUM	50	79,000.00	3,950,000.00
9.	**SAC Code:** **WATERPROOFING:** 1. The waterproofing treatment shall be got executed only through a specialized agency to be approved by the owner, under the supervision of the main contractor. Specialized agency, through which the contractor intends to get the water proofing work done, sholud have completed at least three similar works with satisfactory completion certificates and performance. At least one of these three works should have been completed by the specialized agency, at least five years prior to the date of the tender. 2. Signing guarantee bond as per approved performa for waterproofing treatment for 10 years from the date of final completion on non judicial stamp paper. The guarantee bond shall be executed & extended by the contractor to the client and not by the water proofing agency. The guarantee bond enclosed with tender paper shall be signed by waterproofing agency and the main civil contractor. 3. The rate shall be inclusive of the work in narrow widths, junctions Including leads and lift upto terrace level unless explicitly specified otherwise. 4. Terrace - Providing 50x50mm, size chase in wall, parapets etc. at 300mm height from floor level for tucking ends of vertical layers of the treatment, filling up the chase with cement mortar 1:4 (1 Cement : 4 Course sand) and				

	preparing a drip mould as per detailed drawing just above the chase. 5. The contractor shall get water tanks tested against any leakage/seepage by filling all the water tanks with water for a minimum period of 15 days, including cost of water, all necessary arrangements required for filling and emptying the tank after testing. The testing, rectification of leakage and seepage if any shall be completed to the entire satisfaction of the owner, who may desire the retesting as detailed above, till satisfactory results are obtained, to the entire satisfaction of the client. 6. For measurements, the actual treated area shall be measured. 7. Quoted rates are applicable for all depths, heights, lead & lift unless specified explicitly. 8. Contractor to deposit all the water proofing material(s) in owner's stores and a copy of the bill(s) of such materials to be submitted to the owner. Empty bags/drums/containers of waterproofing material shall be removed only after the written permission of the owner. 9. The Contractor shall submit the daily report of the status of waterproofing material received/consumed and the balance at site, to the client.				
9.1.	**SAC Code:** **TERRACE WATERPROOFING:** Providing and laying 120 mm average thickness of brick bat Coba and 65 mm at Khurra on Terrace through an approved agency, approved by the Architect as per steps given. (a) Surface Preparation, Cleaning the mother slab, repair of joints, Providing and applying URP with cement slurry as primer to fill all the pores etc on mother slab over primer Providing and applying two coats of Fast Flex @ 2kg/Sqm including sandwiching the layer with fiber mesh of 45 GSM between two layers. (It shall also be execute up to 300 mm height of wall) (b) laying cement concrete using broken bricks/ brick bats of size 25 mm to 100 mm size with 50 % of cement mortar 1:5 (1 cement : 5 coarse sand) admixed with proprietary water proofing compound conforming to IS:2645 over 20 mm thick layer of cement mortar of mix 1:5 (1 cement : 5 coarse sand) admixed with admixed with proprietary water proofing compound	SQM	1000	1,150.00	1,150,000.00

	conforming to IS:2645 to required slope and treating similarly the adjoining wall upto 300 mm height including rounding of junctions of wall and slabs. (c) After two days of proper curing applying a second coat of cement slurry admixed with proprietary water proofing compound conforming to IS:2645 (d) Finishing the surface with 20 mm thick jointless cement mortar of mix 1:4 (1 cement : 4 coarse sand) admixed with proprietary water proofing compound conforming to IS:2645 and finally finishing the surface with trowel with neat cement slurry and making 300X300 mm square. (e) The whole terrace so finished sshall be looded with water for a minimum period of two weeks for curing and for final test. All above operations to be done in order and as directed and specified by the engineer in charge. With average thickness of 100 mm thickness and minimum thickness at khurra as 65 mm. Providing gola in 1: 2: 4 cement concrete with 12 mm nominal size stone ballast, finished with cement plaster 1:2 (1 cement: 2 medium sand) laid, cured complete as per design. (Approx. Size 100mm x 100mm) Making khurras 450 x 450 mm with average min. thk. of 50 mm over PVCsheet of 1m x 1m x 400 micron with cement concrete 1:2:4 (1 cement: 2 coarse sand: 4 graded stone aggregate of 12 mm nominal size) finished with 12 mm cement plaster 1:3 (1 cement: 3 coarse sand) and a coat of neat cement, rounding the edges, making and finishing complete. **(Only plan area shall be paid).**				
	Total				**16,20,74,375.00**

TERMS & CONDITIONS:-

A. Agreement Term:-

1. Work will be carried out as per the instructions of Site In charge. The quantities given in the schedule will be verified by you as per site conditions, as per drawings given by the consultants.
2. You will keep in touch with Project Manager at site to decide the priorities work. You shall take the approval of the sample from Project Manager.
3. For your tools tackles and equipments brought at site, you will get the same entered on the gate with your delivery notes so that you can be allowed to take out the same on completion of the work.
4. Variation of quantities may be up to any extent.
5. No work shall be sublet without getting in writing confirmation from undersigned.

6. If any complaint received from Engineer In charge / Consultant during the execution or after handing over of job or in DLP of 12 months same shall be get rectified by you at your own cost or by us at your cost.
7. **Reconciliation of RMC and reinforcement to be strictly submitted with each R/A bill.**
8. The Vendor shall submit the quality test reports with each lot from authorized lab as per IS Code norms for the material to be used. The project-in-charge shall take required random samples from the materials used and get it tested from 3rd Party Lab. In case any material is found to be of inferior quality in accordance with IS Code requirements, suitable action shall be taken against the Vendor.

B. Legal Obligations:-

1. You are fully responsible for any mishapening at your work site. Contractor labour and workers shall follow strictly the safety norms as enclosed herewith and contractor shall be fully responsible for their safety.
2. No worker less than 18 years shall be deployed at site.
3. Your labours and workers shall follow strictly the safety norms as specified and you shall be fully responsible for their safety.
4. You shall keep the labour colony area clean and hygienic. No medical bill shall be paid for treatment of any of the workmen.
5. You shall submit receipt muster roll of your workmen for the previous payments along with his bill of the next period, failing which no further payment shall be made to you. Under all circumstances you shall be solely responsible for the correct and timely payments of your workers.
6. You will submit the PF and ESI deposit challan, if the same is not deposited, shall be deducted from your bill.
7. Contractor will strictly follow NGT guidelines/Local bodies compliances/Liasioning and whole sole responsibilities lies with concerned contractor only. Client is not liable for any non-compliances in this regard by contractor.
8. Labour cess shall be borne by OPRPL.

C. Payment Terms:-

1. **MOBILISATION ADVANCE:-** Rs.60.00 Lakhs shall be paid against undated counter guarantee cheque as under as per below. (Signature should be duly verified by the bank of the same account).
   - Rs.20.00 Lakhs along with order.
   - Rs.20.00 Lakhs on start of Raft concreting.
   - Rs.20.00 Lakhs on bringing the shuttering material for Ground Floor Slab.

   The above advances shall be recovered from R.A. Bills on prorata basis from 2nd R/A bill onwards, however full recovery of advance shall be made upto 80% of work done.
2. **Mobilisation Period:** Contractor shall mobilize within 10 days of issuance of work order.
3. **MODE OF PAYMENT:-** Contractor shall prepare and submit the bills along with all required documents once in a month at site. 70% adhoc payment will be released within 12 days from date of certification of bill duly verified by project manager and balance payment shall be released within next 15 days (including certification from H.O.)
4. **SECURITY DEPOSIT:-** 05% of the gross value of the actual work done shall be deducted from contractor running bills subject to maximum upto 2.5% of contract value. 50% Security deposit will be released along with Final Bill & Balance 50% after Defect Liability Period of twelve months.

5. Payment (including GST) of RMC are to be made directly by the client to the RMC supplier on behalf of the contractor against request letter by contractor after verification and approval of RMC rates and payment terms through client's Purchase Department and the same shall be recovered from very next R.A. Bill of contractor.
6. **Secured Advance:** 75% of material value or basic rate (whichever is less without GST) shall be paid on Structural Steel and Wooden Door Frames on delivery at site as secured advance, the recovery of secured advance shall be done in R.A. Bills on pro-rata basis. The above material shall be consumed within 75 days.

D. Rates:-

1. Any kind of financial implications shall be got approved in writing from the undersigned. Any commitment made by any other representative shall not be binding on OPRPL.
2. Your rate shall remain firm during the currency of the contract or extended period if any. **No escalation whatsoever shall be payable on any account till completion of work except, material basic rate variation as specified in work order.**
3. The above rates are inclusive of all material (except Reinforcement Steel & Hoop Iron), labour, tools, tackles and all consumables stores, transportation, octroi and unloading.

E. Taxation Term:-

1. Only GST shall be paid extra. The Contractor will issue proper GST invoice so that Company can take input credit. The Contractor will file all due returns covers under GST & pay due tax on time. In case the Company face problems to availing input tax credit, the Contractor will indemnify to the Company.
2. ITDS shall be deducted as per applicable rates.

F. Special Terms:-

1. The Basic Rate of following materials are FOR Site but excluding GST.
   - **Bricks @ Rs.7.00 per No.**
   - **PCC M-10 @ Rs.4200/- per Cum without pump.**
   - **RMC M-25 @ Rs.5200/- per Cum without pump. For higher grade RMC including SCC - Only basic rate difference shall be payable.**
   - **Structural Steel @ Rs.60/- per Kg.**

   Further only difference in Basic Rate shall be paid/recovered subject to prior approval and with the consent of Client Purchase Department.
2. **Basic rate shall be considered only for Bricks, Structural Steel, PCC and RMC as mentioned above. Nothing shall be considered for any hike in price for any other material except mentioned above.**
3. **Any variation (upward/ downward, both cases) in basic rate versus purchase rates shall be paid / recovered, only if prior approval taken by contractor through client's procurement department.**
4. **Contractor shall submit monthly/ fortnightly or as per requirement of material at site (well in advance) to site team for approval of material purchase rates through client's procurement department before procurement of materials, this is applicable for which base rate is mentioned in work order.**
5. **For the purpose of basic rate variation (+/-), the billed quantities as per approved drawing will be considered for basic rate variation unless otherwise mentioned in work order.**
6. Construction as well as Drinking water shall be in the scope of contractor including its further

distribution and storage.

7. Electricity for construction and labour colony shall be in the scope of contractor including perimeter lighting. Electricity shall be provided at one point (if available) on chargeable basis further distribution is in the scope of contractor. Nothing extra shall be paid on this account whatsoever.
8. Only space shall be provided by Client for labour hutment, further construction of labour hutment, sanitation, house keeping, water, electricity arrangement shall be made by contractor at his own cost.
9. **Rates are including for all lead and lift. Nothing extra shall be paid on any account towards any lead, lift or any additional height/depth.**
10. **All Elevation features shall be paid in given BOQ items. Nothing shall be paid extra for any elevation feature work on any account whatsoever.**
11. Scope of work can be increase or decrease at any stage and no claim shall be entertained on any account whatsoever.
12. **INSURANCE:-** Third party insurance shall be taken by contractor.
13. **SECURED ADVANCE:-** Refer clause no.C-6..
14. Workmen Compensation Policy shall be taken by contractor.
15. Watch & Ward of the premises will be in the scope of contractor & proper barricading will be in the scope of contractor.
16. All wok will be executed as per latest CPWD specification, IS Code and as directed by EIC.
17. Nothing shall be paid for any claim for any hindrances/stoppage of work due to any Govt. order or any natural disaster etc. No idle charge shall be paid in any case.
18. **If there is Any Change (Plus or Minus) in Tax Provision by the Government either State or Central, the same shall be paid or recovered as applicable.**
19. Good safety net shall be provided by the contractor as per direction of EIC.
20. Functional QA/QC lab shall be established by the contractor to maintain the required quality at the project for meeting the quality standards and contractor shall depute a proper Q.A. Engineer from the beginning and submit a quality assurance plan for the same.
21. Contractor shall submit to the Project Manager, drawings giving his proposed layout of locating offices, stores, godowns, yards, water and electricity network etc. for approval of EIC.
22. Shuttering material, scaffolding detailing is to be submitted to P.M./EIC minimum 15 days prior to carry for approval, who may seek opinion of the Structural Consultant.
23. Contractor shall submit Organization Chart and work completion schedule within 15 days of issue of work order for approval of EIC.
24. Contractor shall make his own arrangements/tube well for construction water including testing, storage and distribution of water.
25. Maintaining good quality standards as per satisfaction of our client/EIC with timely achievement of the set milestones at this project.
26. 1 No. Mobile crane & 1 No. Material Hoist shall be in the scope of contractor.
27. DG of required capacity for Mobile Crane and Material Hoist is in the scope of contractor.
28. Survey work is in the scope of contractor. Client personnel will verify it.
29. Safety net minimum at 2 level shall be provided by contractor.
30. HDPE Debris chute is in contractor scope.
31. Surface water & rainy water pumping out is in contractor scope and included in item rate, nothing shall be paid extra.

G. General Terms:-

1. We reserve the right to terminate the contract at any stage without assigning any reason. You will not claim any compensation for this.
2. All disputes are subject to New Delhi jurisdiction.
3. **FREE ISSUE MATERIALS:-** Not withstanding anything contained at any other clause/s, the contract value is excluding the cost of following material, however installation for the same shall be in contractor scope:

Material	Maximum permissible wastage
Reinforcement Steel & Hoop Iron	2.5%

Reconciliation of RMC, Cement & client free issued materials **(Reinforcement & Hoop Iron)** shall be carried out in each Running bills & Final Bill. If wastage of free issued materials **(Reinforcement & Hoop Iron)** exceeded the above specified limit, Cost of the excess wastage quantity shall be recovered from the Contractor on the basis of 1.5 times of actual cost to client. Cost of reinforcement if used for shuttering and other than scope of structural drawings, the same shall be recovered @ 1.5 times of actual cost to client. The decision on the actual cost given by the Client shall be final and binding on the Contractor. All other material except FOC's shall be provided by Contractor at his own cost.

Please sign with stamp the duplicate copy of this order in token of your acceptance.

Thanking you,

Yours Faithfully
For **Omaxe Pancham Realcon Private Limited**

Received & Accepted
For MARCO ENTERPRISES

Mr. Sanjay Gupta	**Mr. Amit Singal**	**Mr. Rakesh Gupta**	**Mr. Ravi Chauhan**
(President Contracts)	**(V.P. Contracts)**	**(V.P. Construction)**	**(Addl. V.P. Projects)**

Authorised Site Contact Person:	**Mr. Suresh Kumar Pandey**
Designation:	**G.M. (Construction)**
Mobile No:	**9711800470**
Email ID:	**sureshkumarpandey@omaxe.com**

Encl:- **1. Safety Policy (06 Pages)**
2. Steel Specification (08 Pages)

Please submit your Invoice at Email id invoice@omaxe.com

GENERAL TERMS & CONDITIONS

1. **Suspension of work:** Client may at any time, in its sole discretion, immediately suspend the execution of work or part thereof. During suspension of works or part thereof the contractor shall continue to perform the work that are not suspended.

2. **Rate of Delay Liquidated Damages:** 0.5% of the Contract Price for every week of delay prorated to the actual number of days of delay. Maximum Limit of Delay Liquidated Damages 5% of the Contract Value.

3. **FORCE MAJEURE:** Force Majeure means exceptional event of circumstance

   a) Which is beyond a Party's Control

   b) Which such party could not reasonable have provided against before entering into the Contract,

   c) Which, having, arisen, such party could not reasonably have avoided or overcome, and

   d) Which is not substantially attributable to the other party.

   Force Majeure may include, exceptional events or circumstance

   i) War, hostilities (whether war be declared or not), invasion, act of foreign enemies,

   ii) Rebellion, terrorism, revolution, insurrection, military or usurped power, or civil war,

   iii) Riot, commotion, disorder, strike or lockout by persons other than the Contractors Personnel and other employees of the Contractor.

   iv) Munitions of war, explosive materials, ionizing radiation or contamination by radio-activity, except as may be attributable to the Contractor's use of such munitions, explosives, radiation or radio-activity, and Natural catastrophe such as earthquake, hurricane, typhoon or volcanic activity.

4. **Return of surplus materials:** Notwithstanding anything contained to the contrary in this contract, where any materials for the execution of the contract are procured with the assistance of client either by issue from Client stocks or purchase made under orders or permits or licenses issued by Client, the contractor shall hold the said materials economically and solely for the purpose of the contract and not dispose of them without the written permission of the client and return, if required by the Project-in-Charge, all surplus or unserviceable materials that may be left with him after the completion of the contract or at its termination for any reason whatsoever on being paid or credited such price as the Project-in-Charge shall determine having due regard to the condition of the materials. The price allowed to the contractor however shall not exceed the amount charged to him excluding the element of storage charges. The decision of the Project in Charge shall be final and conclusive. In the event of breach of the aforesaid condition, the contractor shall in addition to throwing himself open to action for contravention of the terms of the license or permit and/or for criminal breach of trust, be liable to Client for all moneys, advantages or profits resulting or which in the usual course would have resulted to him by reason of such breach.

5. **Carrying out part work at risk & cost of contractor:** If contractor:

(i) At any time makes default during currency of work or does not execute any part of the work with due diligence and continues to do so even after a notice in writing of 7 days in this respect from the Project-in-Charge; or

(ii) Commits default in complying with any of the terms and conditions of the contract and does not remedy it or takes effective steps to remedy it within 7 days even after a notice in writing is given in that behalf by the Project-in-Charge; or Fails to complete the work(s) or items of work with individual dates of completion, on or before the date(s) so determined, and does not complete them within the period specified in the notice given in writing in that behalf by the Project-in-Charge.

The Project-in-Charge without invoking action under other clauses of contract, without prejudice to any other right or remedy against the contractor which have either accrued or accrue thereafter to client, by a notice in writing to take the part work / part incomplete work of any item(s) out of his hands and shall have powers to:

(a) Take possession of the site and any materials, constructional plant, implements, stores, etc., thereon; and/or

(b) Carry out the part work / part incomplete work of any item(s) by any means at the risk and cost of the contractor. The Project-in-Charge shall determine the amount, if any, is recoverable from the contractor for completion of the part work/ part incomplete work of any item(s) taken out of his hands and execute at the risk and cost of the contractor, the liability of contractor on account of loss or damage suffered by Client because of action under this clause shall be as per actuals. In determining the amount, credit shall be given to the contractor with the value of work done in all respect in the same manner and at the same rate as if it had been carried out by the original contractor under the terms of his contract, the value of contractor's materials taken over and incorporated in the work and use of plant and machinery belonging to the contractor.

The certificate of the Project-in-Charge as to the value of work done shall be final and conclusive against the contractor provided always that action under this clause shall only be taken after giving notice in writing to the contractor. Provided also that if the expenses incurred by the department are less than the amount payable to the contractor at his agreement rates, the difference shall not be payable to the contractor.

Any excess expenditure incurred or to be incurred by Client in completing the part work/ part incomplete work of any item(s) or the excess loss of damages suffered or may be suffered by Client as aforesaid after allowing such credit shall without prejudice to any other right or remedy available to Client in law or per as agreement be recovered from any money due to the Contractor on any account, and if such money is insufficient, the contractor shall be called upon in writing and shall be liable to pay the same within 30 days.

If the contractor fails to pay the required sum within the aforesaid period of 30 days, the Project-in-Charge shall have the right to sell any or all of the Client unused materials, constructional plant, implements, temporary building at site etc. and adjust the proceeds of sale thereof towards the dues recoverable from the contractor under the contract and if thereafter there remains any balance outstanding, it shall be recovered in accordance with the provisions of the contract.

In the event of above course being adopted by the Project-in-Charge, the contractor shall have no

claim to compensation for any loss sustained by him by reason of his having purchased or procured any materials or entered into any engagements or made any advance on any account or with a view to the execution of the work or the performance of the contract.

6. **SEVERABILITY:** If any term or provision of the Contract or the application thereof to any person or circumstances shall, to any extent and for any reason, be invalid or unenforceable, the remainder of the Contract, or the application thereof to persons or circumstances other than those as to which it is held invalid or unenforceable, shall not be affected thereby and each term and provision of the Contract shall be valid and be enforced to the fullest extent permitted by law.

7. **SURVIVAL:** Rights and obligations accrued prior to expiry or termination of the Contract and all rights and obligations, which by their nature are intended to survive the expiry or termination of the contract, shall survive such expiry or termination.

8. **INSPECTION OF SITE:** The Contractor shall be deemed to have inspected and examined the site and its surroundings and information available in connection therewith and to have satisfied himself, before accepting the Contract, as to the form and nature thereof, including the sub-surface conditions, the hydrological and climatic conditions, the extent and nature of work and materials necessary for the completion of the Works and the remedying of any defects therein, the means of access to the Site and the accommodation he may require and, in general, shall be deemed to have obtained all necessary information, as to risk, contingencies and all other circumstances which may influence or affect this Contract and the Client shall not in any event be liable for any damage or loss caused to the Contractor in relation to the same.

SAFETY POLICY

(A) ADMINISTRATIVE CONTROL

- Safety officer, who will implement the safety procedure, should be available at the site.

- First aid box to be kept at following points:
  a) at Guard room
  b) at Project Engineer`s office
  c) at Contractors Site-in-Charge office
  d) at store.

- Every labour, supervisor, engineer or any other person at site should wear helmet properly with tight support. Helmets should have identification by way of company name/ logo and color. Suggested colors are as follows:

  a) For Project I/C, PM & HO senior staff (During visit at site) : BLUE
  b) For Safety Dept. Rep. : RED
  c) Foreman, Supervisor : GREEN
  d) Workmen : YELLOW
  e) External Visitors at site : WHITE

- Any person entering to project site should wear shoes/ safety shoes (if required).

- Safety belt should be used by labourers doing work at heights. Safety belt should also be used by Engineers & Supervisors during execution supervision/ inspection at height.

- All opening in the slab should be covered with temporary welded 8mm steel mesh @ 6' c/c spacing & covered with nylon mesh.

- All edge side of slab, sides of waist slab of staircase should be barricaded by 8mm steel welded temporarily.

- Nylon safety net should be tied alround structure just at below level slab, where work is being done.

- Education for observing safety rules should be given regularly by safety officer to skilled/unskilled labourers and supervisory staff at site.

- Telephone no. of nearby hospital & responsible person of agency should be displayed at prominent spots at site, where it is in view of all persons.

- Security guard should not allow anyone to enter in the site without following the safety prosecutions.
- Insurance of labourers should be ensured.
- Crèche for children should be made at a safe place including deployment of maid servant capable to taking care of children.
- Proper slope/offset should be maintained in earth cutting.
- Proper barricading for entire boundary at the construction site should be done.
- No slant props in shuttering should be allowed.
- No weak member in shuttering should be allowed.
- Site in charge of agency should satisfy him self about strength of centering & shuttering before allowing casting of slab.
- No loose brick should be left on edge of slab/balcony.
- All deep excavations should be protected by proper barricading and warning sign should be displayed near such spots.
- Working platform, gangways, stairways shall be built not to sag having proper width and fencing.
- Use of bamboo is to be strictly prohibited for any construction activity at site.
- Steel scaffolding shall be provided for workmen for all work done at site.
- No portable single ladder shall be over 5 meters in length. Width between side rails in a rung ladder shall in no case be less than 30 cm. for ladders up to and including 3 m. in length. For longer ladders this width shall be increased at least 16 mm for each additional 30 cm. length. Uniform step spacing shall not exceed 30 cm.
- No materials on the site shall be so stacked or placed as to cause danger or inconveniences to any person or the public.
- Every rope used in hoisting or lowering materials or as a means of suspension shall be of durable quality and adequate strength.
- All trenches 1.2 metres or more in depth shall at all times be supplied with at least one ladder for each 30 metres in length or fraction thereof. Ladder shall be extended from bottom of the trench to at least 1 metre above the surface of the ground.

- Earth Cutting shall be done from top to bottom. Under no circumstances undermining or under cutting shall be done.

- Scrap must be removed from construction site and disposed off promptly.

- Packaging material should be disposed of quickly and must not be allowed to remain at construction site. Packaging material presents fire hazard.

- Any person suffering from HEIGHT PHOBIA should not be allowed to work at height.

- Wearing of Kurta, Payjama, Muflor or any kind of loose cloth should not be allowed to work at height.

- A record register should be maintained at site, in which name, permanent/ local address with telephone number & one photograph of all the persons working at site with name, address and telephone numbers of any relative or friend in case of emergency. This record should be in custody of Project I/C or PM (Responsible person generally available at site).

- No temporary tea stall, dhaba of khoka shall be allowed in the premise of construction area. It should at least be 50 mtr away from site.

(B) SERVICES CONTROL

- Fire fighting equipment in working condition should be available at site at approachable locations.

- An additional person should be deployed to hold ladder, if ladder is necessary at any point.

- Electric wires should be of proper insulation & should pass underground or above ground over poles. Electrical wires should not allowed to travel underground as mentioned. In case underground route is required, we should use aluminium conductor armaured cable.

- No joint of electric wire should be open.

- Meter box/junction box should be covered & protected.

- Proper lighting of entire site by means of flood light be ensured.

- Licenced plumber/electrician should be deployed for their respective jobs.

- Proper servicing of all mechanical means being used in construction work should be ensured.

- Electric wires are properly insulated and frequently checked for the safety of worker at site, installation of MCB at junction to be insured so that over loading can be avoided.

- Power distribution system shall be identifiable with display marking on all switches.

- Meter room and main switches should be freely accessible at all times and fully protected against all weather.

- Motors, gearing transmission, electrical wiring and other dangerous parts of hoisting appliances shall be provided with efficient safeguards.

- Workers employed on electrical installations which are already energized, including mats shall wear apparel such as gloves, sleeves and boots.

- Workers shall not wear any rings, watches and keys or other materials which are good conductors of electricity.

- All power distribution wiring shall be carried out concealed, adequately insulated and of appropriate current/load rating cables. It shall be securely routed for the purpose. No loose, naked, hanging wire shall be permitted.

- Ear – Masks for protection of hearing system from loud sound.

- A Feeder Piller should be installed at site for construction power distribution work. All the electrical connections should be taken from this Feeder Piller.

- Proper earthing should be done for electro – mechanical tools & equipments working at site by means of PVC insulated wire, flexible GI rope or GI strip.

- All workers should be equipped with proper tools during work.

- Tools required for electrical work i.e. pliers, screw drivers etc. should be properly sleeved.

- Use of hoisting machines and tackle including their attachments, anchorage and supports shall confirm to the following standard or conditions :-

  (a) These shall be good mechanical construction, sound material and adequate strength and free patent defect and shall be kept in good repairs and in good working order.

  (b) Every rope used in hoisting or lowering materials or as means of suspension shall be of durable quality and adequate strength, and free from patent defects.

(c) Every crane driver or hoisting appliance operator shall be properly qualified and no person under an age of 21 years should be in – charge of any hoisting machine including any scaffold which or give signals to the operator.

(d) In case of every hoisting machine and of every chain ring hook, shackle swivel and pulley block used in hoisting or lowering or as means of suspension the safe working load shall be ascertained by adequate means. Every hoisting machine and all gear referred to above shall be plainly marked with the safe working load. In case of hoisting machine having a variable safe working load, each safe working load of the conditions under which it is applicable shall be clearly indicated. No part of any machine or of any gear referred to above in this paragraph shall be loaded beyond the safe working load except for the purpose of testing.

(C) GENERAL AWARENESS

- Site up keep should be followed and in honest way.
- A routinal exercise should be done at site to identify all weak points of safety & their remedial measures.
- Visitors shall be allowed to visit site only accompanied by safety officer or representative.
- Agency shall ensure that ambulance service is readily available in case of any emergency.
- Smoking within site, particularly near lubricant/flamable objects should be strictly prohibited.
- These safety provisions shall be brought to the notice of all concerned by display on a notice board at a prominent place at the site.
- When work is done near any place where there is risk of drowning all necessary rescue equipment shall be provided and kept ready for use and all necessary steps taken for prompt rescue of any person.
- Workers employed on mixing asphaltic materials, cement and lime mortars/concrete shall be provided with protective footwear and protective gloves.
- Those engaged in handling any materials which are injurious to eyes shall be provided with protective goggles.
- Those engaged in welding works shall be provided with welders protective eye shields.

- Stone breakers shall be provided with protective goggles and protective clothing and seated at sufficiently safe spacing.

- Workers employed on concrete finishing, welding, painting and other works above 2 meters shall be provided with a suitable safety belt.

- When workers are employed in sewers and manholes, which are in use, the contractor shall ensure that manhole covers are opened and manholes are ventilated at least for an hour before workers are allowed to enter them.

- No one should be deployed below the age of 18.

- No paint containing lead or lead products shall be used except in the form of paste of ready-made paint.

- Suitable face masks shall be supplied for use by workers when the paint is applied in the form of spray or a surface having lead paint is dry rubbed and scraped.

- Proper space for shuttering yard, steel yard, cement store, DG area should be allocated at site.

- Safety glass for welders should be ensured.

- Routine health check up should be done.

- A regular review of safety aspects should be done & maintained in a proper format should be available for checking at site.

- SAFETY BOARDS (as per approved sample) shall be fixed at the proper location/s inside site premise. (Quantity as required).

- Before any demolition work is commenced and also during the demolition –

  (a) All roads and open areas adjacent to the work site shall either be closed or suitable protected

  (b) All practical steps shall be taken to prevent danger to persons employed from risk or fire or explosion or flooding. No floor, roof or other of the building shall be so over – loaded with debris or materials as to render it unsafe.

CONDITIONS FOR REINFORCEMENT STEEL (ISSUED FREE OF COST)

The Reinforcing Steel shall be procured by the Owner, at its own expenses and issued to the Contractor free of cost for consumption in the work, unless specifically stated otherwise in the "Description of work" in the itemised schedule of quantities.

Reinforcement steel as supplied to the CONTRACTOR shall remain the absolute property of the OWNER and shall not be removed on any account from the SITE of the WORK, and shall be at all times open for inspection to the EIC. Any such materials remaining unused at the time of completion or termination of the CONTRACT shall be returned to the OWNER's stores or at a place as directed by the EIC in perfectly good condition at CONTRACTOR's cost.

1. Conditions for issue of reinforcement steel

a) Steel to be issued by the OWNER will be supplied to the CONTRACTOR from his stores or at site of work. It shall be responsibility of the CONTRACTOR to take delivery of the materials and arrange for its loading, transportation and unloading at the SITE of WORK at his own cost. The materials shall be issued between the working hours and as per the rules of the OWNER as framed from time to time.

b) The CONTRACTOR shall bear all incidental charges for the storage and safe custody of materials at site after these have been issued to him.

c) Steel specified as to be issued by the OWNER shall be issued in standard sizes as obtained from the manufactures.

d) The CONTRACTOR shall construct suitable stores at the SITE of work for storing the materials safe against damage by rain, dampness, fire, theft etc. He shall also employ necessary watch and ward establishment for the purpose.

e) It shall be duty of the CONTRACTOR to inspect the steel supplied to him at the time of taking delivery and satisfy himself that they are in good condition. After the materials have been delivered by the OWNER, it shall be the responsibility of the CONTRACTOR to keep them in good condition and if the materials are damaged or lost, at any time, they shall be repaired and / or replaced by him at his own cost according to the instructions of the Architect/Owner/Engineer-in-Charge.

f) The OWNER shall not be liable for delay in supply or non-supply of steel which the OWNER has undertaken to supply where such failure or delay is due to events of force majeure such as natural calamites, act of enemies, transport and procurement difficulties and any circumstances beyond the control of the OWNER. In no case, shall the CONTRACTOR be entitled to claim any compensation or loss suffered by him on this account.

g) None of the steel supplied to the CONTRACTOR will be utilized by the CONTRACTOR for manufacturing items which can be obtained as supplied from standard manufacturer in finished form.

The CONTRACTOR shall, if desired by the EIC, be required to execute an Indemnity Bond in the prescribed form for safe custody and accounting of all materials issued by the OWNER.

h) The Contractor shall furnish to the EIC sufficiently in advance a statement showing his requirement or the quantities of the steel to be supplied by the Owner and the time when the same will be required by him for the works, so as to enable the EIC / EIC to make necessary arrangements for procurement and supply of the same.

i) Account of the steel issued by the Owner shall be maintained by contractor indicating the daily receipt, consumption and balance in hand. This account shall be maintained in a manner prescribed by the EIC along with all connected papers viz. requisitions, issues, etc. and shall be always suitable and available for inspection in the Contractor's office at Site.

j) The Contractor should see that only the required quantities of steel are got issued. The Contractor shall not be entitled to cartage and incidental charges for returning the surplus steel, if any, to the stores where from they were issued or to the place as directed by the EIC.

k) Steel supplied by Owner shall not be utilized for any purpose(s) other than issued for.

l) The contractor will have no right of whatsoever nature, in the material issued to him (for the consumption in the work) and all the rights in the property of materials will vest only with the Owner. All materials supplied by the Owner shall be used by the Contractor exclusively for the construction work for which they are issued and on no account the material shall be taken out of the Owner's premises by the Contractor without prior and due permission.

m) Reinforcement bars, shall be supplied by weight. However the Contractor shall ensure that the steel supplied to him is of exact dia/thickness/ size/ weight as per relevant IS code. The materials will be supplied in such lengths and sizes as procured by the Owner / received at site.

n) The Contractor's rates for labour / laying / fixing for the items involving consumption of materials supplied by the Owner shall cover the cost of collecting, quantity / measurement checking and transporting the material from the place of delivery to site including cost of unloading the materials from the trucks to the contractor's stores or site of work as also the cost of returning the materials left surplus and in good condition as acceptable to Owner after the completion of the work to the Owner's stores or to the place at site as directed by the Owner. The Contractor shall be responsible for the safety and proper storage of all the materials supplied by the Owner.

o) In case wastage, expressed as percentage of the theoretical consumption exceed the permissible limits as indicated below. All except unaccounted wastage of Mild/Tor/TMT steel will be the property of the Owner.

Materials	**Permissible Wastage**
Reinforcement steel bars TMT bars of all diameters	2.5% excluding rolling margin

The excess ABOVE 2.5% wastage will be recovered at the rates as below:-

The rolling margin shall be recorded with each vehicle and shall be signed off by client for admissibility, if contractor fails to do so, rolling margin will not be considered.

Permissible wastage maximum 2.5% excluding rolling margin, Recovery Rate 1.50 times beyond permissible wastage on average procurement rate.

p) In case of steel items, wastage shall include unaccounted wastage due to use of steels of larger diameter as well as sizes and quantities used in excess of those shown in the drawings as also steel cut pieces of sizes as stated below:-

i) Cut pieces of reinforcement of steel as below shall be considered as wastage.

a) Upto 12mm dia - Length below 2 meter

b) 16mm & above dia - Length below 2 meter

Note: All reinforcement steel shall be returned (except wastage defined above) in good usable condition.

q) The contractor shall give requirements of steel after proper workings and the additional required in writing to the Engineer-in-Charge minimum four weeks in advance. In case material supplied by the Owner as per contractor's written request surplus becomes surplus the Contractor shall return such surplus materials at the place of issue at his cost.

r) If the materials which are issued to the contractor are lost and / or damaged while in the custody of the contractor and /or damaged or lost in transit or for whatsoever reason after the issue, the cost of such material will be recovered the rates specified.

s) The scrap will be property of Owner and shall be sold by the Employer in consent with the Contractor for reconciliation purpose.

t) The theoretical consumption of steel shall be worked out as follows:-

i) For reinforcement steel, the quantities shall be as per drawings / measurements jointly recorded, duly checked and certified as per I.S.tables.

u) The Contractor shall prepare steel issue slips/statements showing details for quantity of materials issued by the Owner, duly certified by EIC and materials consumption statements of steel reconciled with the measured quantities of works. Both the above statements duly certified by EIC shall be enclosed in every interim bills.

v) Contractor shall inspect the materials and satisfy himself that they are in good conditions at the time of taking materials as required by specifications and / or as instructed by Engineer.

w) Following shall be responsibility of Contractor:

i) Contractor shall check the quantity of supplied material and any defects like hair cracks, spots etc. must be brought to the notice of Owner.

ii) Materials must be insured for any theft, loss, damage etc. by the contractor at his own cost, while in custody of contractor.

iii) In case contractor fail to get the materials insured, Owner shall do the same and recover the cost from the Contractor.

iv) It shall be sole responsibility of contractor to make stores for stocking of material & safe custody.

v) Mishandling etc. shall be contractors responsibility.

vi) Reconcillation of steel in each bill.

vii) Reconcillation to include Quantity taken by Contractor from client. Quantity consumed in actual works, as per drawings & balance in stock. Difference to be explained to Owner and accepted by Owner.

TECHNICAL SPECIFICATIONS

1. REINFORCEMENT

1.1 Assembly of Reinforcement

Bars shall be bent correctly and accurately to the size and shape as shown in the detailed drawing or as directed by Engineer-in-Charge. Preferably bars of full length shall be used. Necessary cutting and straightening is also included. Over lapping of bars, where necessary shall be done as directed by the Engineer-in-Charge. The overlapping bars shall not touch each other and these shall be kept apart with concrete between them by 25mm or $1^1/4$ times the maximum size of the coarse aggregate whichever is greater. But where this is not possible, the overlapping bars shall be bound together at intervals not exceeding twice the dia of such bars with two strands annealed steel wire of 0.90mm to 1.6mm twisted tight. The overlaps/splices shall be staggered as per directions of the Engineer-in-Charge. But in no case the over lapping shall be provided in more than 50% of cross sectional area at one section.

1.2 Bends and Hooks Forming End Anchorages.

Reinforcement shall be bent and fixed in accordance with procedure specified in IS 2502, code of practice for bending and fixing of bars for concrete reinforcement. The details of bends and hooks are shown below for guidance.

a) U - Type Hook

In case of mild steel plain bars standard U type hook shall be provided by bending ends of rod into semicircular hooks having clear diameter equal to four times the diameter of the bar.

Note : In case of work in seismic zone, the size of hooks at the end of the rod shall be eight times the diameter of bar or as given in the structural drawing.

b) Bends

Bend forming anchorage to a M.S. plain bar shall be bent with an internal radius equal to two times the diameter of the bar with a minimum length beyond the bend equal to four times the diameter of the bar.

1.3 Anchoring Bars in Tension

Deformed bars may be used without end anchorages provided, development length requirement is satisfied. Hooks should normally be provided for plain bars in tension. Development length of bars will be determined as per clause 25.2.1 of IS:456.

1.4 Anchoring Bars in Compression

The anchorage length of straight bar in compression shall be equal to the 'Development length' of bars in compression as specified in clause 25.2.1 of IS : 456. The projected length of hooks, bends and straight lengths beyond bend, if provided for a bar in compression, shall be considered for development length.

1.5 Binders, stirrups, links and the like : In case of binders, stirrups, links etc. the straight portion beyond the curve at the end shall be not less than eight times and nominal size of bar.

1.6 Welding of Bars

Wherever facility for electric arc welding is available, welding of bars shall be done in lieu of overlap. The location and type of welding shall be got approved by the Engineer-in-charge. Welding shall be as per IS : 2751 for mild steel bars and for cold worked bars.

2. Placing is Position

2.1 Fabricated reinforcement bars shall be placed in position as shown in the drawings or as directed by the Engineer-in-Charge. The bars crossing one another shall be tied together at every intersection with two strands of annealed steel wire 0.9 to 1.6mm thickness twisted tight to make the skeleton of the steel work rigid so that the reinforcement does not get displayed during deposition of concrete. Special care has to be taken for the placement of top bars in position as shown in the drawings.

Tack welding in crossing bars shall also be permitted in lieu of binding with steel wire if approved by Engineer-in-Charge.

2.2 The bars shall be kept in correct position by the following methods :

a) In case of beam and slab construction precast cover blocks in cement mortar 1:2 (1 cement :2 coarse sand) about 4x4 cm section and of thickness equal to the specified cover shall be placed between the bars and shutterings, so as to secure and maintain the requisite cover of concrete over reinforcement.

b) In case of cantilevered and doubly reinforced beams or slabs, the vertical distance between the horizontal bars shall be maintained by introducing chairs, spacers or support bars of steel at 1.0 meter or at shorter spacing to avoid sagging.

c) In case of columns and walls, the vertical bars shall be kept in position by means of timber templates with slots accurately cut in them; or with block of cement mortar 1:2 (1 cement : 2 coarse sand) of required size suitably tied to the reinforcement to ensure that they are in correct position during concreting.

d) In case of other R.C.C structure such as arches, domes shells, storage tanks etc, a combination of cover blocks, spacers and templates shall be used as directed by Engineer-in-Charge.

2.3 Tolerance on Placing of Reinforcement

Unless otherwise specified by the Engineer-in-Charge, reinforcement shall be placed within the following tolerances :

Tolerance
in spacing

a) For effective depth, 200 mm or less +/- 10 mm

b) For effective depth, more than 200 mm +/- 15 mm

The cover shall in no case be reduced by more than one third or specified cover or 5mm whichever is less.

2.4 Bending at Construction Joints

Where reinforcement bars are bent aside at construction joints and afterwards bent back into their original position care should be taken to ensure that at no time the radius of the bend is less than 4 bar diameters for plain mild steel or 6 bar diameters for deformed bars. Care shall also be taken when bending back bars to ensure that the concrete around the bar is not damaged.

2.5 Measurement (for reconciliation of theoretical vs actual consumption)

Reinforcement including authorised spacer bars and lappages shall be measured in length of different diameters, as actually (not more than as specified in the drgs.) used in the work nearest to a centimetre and their weight calculated on the basis of standard weight given in Table-4 below. Wastage and unauthorised overlaps shall not be paid for Annealed steel wire required for binding or tack welding shall not be measured, its cost being included in the rate of reinforcement.

Wherever tack welding is used in lieu of binding, such welds shall not be measured. Chairs separators etc shall be provided as directed by the Engineer-in-Charge and measured separately and paid for.

Cross Sectional Area and Mass of Steel Bar

Nominal Size	Cross Sectional Area Sq.mm	Mass per metre Run Kg
6	28.3	0.222
7	38.5	0.302
8	50.3	0.395
10	78.6	0.617
12	113.1	0.888
16	201.2	1.58
18	254.6	2.00
20	314.3	2.47
22	380.3	2.98
25	491.1	3.85
28	616.0	4.83
32	804.6	6.31
36	1018.3	7.99
40	1257.2	9.85
45	1591.1	12.50
50	1964.3	15.42